



PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: KaOne

Version: 1.0

1. Project Overview

- **Problem Statement:** Small business owners and entrepreneurs in Malaysia often struggle to manage finances, predict tax obligations (LHDN) and separate personal from business expenses without expensive accounting software.
- **Target Domain:** AI Financial Intelligence for Malaysian Small Business Owners
- **Proposed Solution Summary:** KaOne is an "AI Pocket CFO" that uses natural language to help users track inventory, forecast budgets with explanation and optimize expenses. It shifts the user's role from a manual bookkeeper to a strategic decision-maker.

2. Background & Business Objective

- **Background of the Problem:** Traditional bookkeeping follows a manual, linear pathway that requires specialized skills. Non-experts often find it tedious to visualize financial health or predict future costs.
- **Importance of Solving This Issue:**
- **Strategic Fit / Impact:** KaOne accelerates financial analysis from hours to seconds. It breaks the technical barrier allowing anyone to manage their business without years of training.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enabled partner in financial management and growth

3.2. Intended Users (Target Audience):

- Individuals
- Freelancers
- Small business owners
- Home-based entrepreneurs

4. System Functionalities

4.1. Description: Manage business through simple conversation. Whether you are checking inventory or forecasting next month's profit, just ask in plain language and let the system do the work.

4.2. Key Functionalities:

- **Inventory update section:** A dynamic table for tracking raw materials with "Add Item" and "Update Stock" capabilities.
- **Budget Planner:** Visual progress bars for categories such as ingredients, marketing, delivery and utilities.
- **Forecasting and Predictions (Gemini Pro-powered):** Generates table showing predicted revenue, expenses and recommended tax savings for the upcoming month.
- **AI Analyst:** Analyse financial by upload document, receipt photo or voice input (Coming soon) and ask anything about finances such as tax predictions, expense breakdown.

- **Cloud Database:** All user data including transactions, inventory items, and split preferences are stored in Google Firebase. This means data persists across browser refreshes and different devices when users log in.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: For the core intelligence of KaOne, we have selected Gemini Pro as our primary Large Language Model.

- This model specifically optimized for the Malaysian financial landscape providing superior accuracy.
- It excels at processing English and Malay language which is critical for accurately parsing unstructured descriptions in Malaysian bank statements and handwritten receipts.
- The model's ability to perform abductive reasoning allows it to logically infer whether an ambiguous transaction is likely a business expense based on the user's profile.

4.3.2. Prompting Strategy: Our team employs a multi-step agentic prompting strategy because financial data is sensitive and requires high precision unlike a single-pass (zero shot) prompt often fails to catch complex patterns in mixed personal/business spending. This strategy fits our use case perfectly because it mimics the workflow of a real human CFO, moving from data gathering to analysis and finally to strategic recommendation

- **Step 1: Intent Extraction:** The first agent identifies the user's intent such as "Analyze tax" vs "Update stock"
- **Step 2: Data Grounding:** The second agent retrieves the specific inventory or transaction data from the user's uploaded history to ground the response in fact and preventing hallucinations.

- Step 3: Logical Validation: A final "reviewer" agent checks the mathematical output such as ensuring predicted tax equals ~18% of income before rendering the final card on the dashboard.

4.3.3. Context & Input Handling: KaOne is designed to handle unstructured financial data such as bank statements CSVs, receipt photos and natural language queries.

- The system utilizes Gemini Pro to parse unstructured text from bank exports or OCR-processed receipt images into structured JSON objects like date, description, amount and category.
- The product accepts

4.3.4. Fallback & Failure Behavior: To ensure accuracy, KaOne incorporates several safety mechanisms:

- If the model returns an unusable or off topic response, the chat interface displays a message, asking the user to provide more specific financial details.
- The system uses a rule-based validator to check if AI-generated tax predictions or budget totals are mathematically sound.
- For database write failures, the system shows a "Save failed" message and allows the user to retry. Data is not lost because the user can re enter the information.

5. User Stories & Use Cases

5.1 User Stories:

1. As a home baker with limited knowledge of accounting, I want a system that can automatically forecast next month's revenue and expenses so that I can plan purchases without needing to do complex calculations myself.
2. As a new street vendor entrepreneur, I want a financial assistant who guides me to plan my budgets with all the receipts and invoices and gives recommendations so that I

can prevent overspending as a new starter, as well as prevent early losses.

3. As a dessert business owner, I want to add new inventory items for new dessert inventions with their costs so that my budget will reflect accurate overall ingredient expenses.

4. As a small business owner, I want to see the visual progress bars of my business income to see if there are improvements in terms of profits and vice versa.

5. As an occupied baker, I want to speak to the system directly to log expenses or stock updates so that I can record information hands-free while I'm baking or shopping.

5.2 Use Cases (Main Interactions):

Use Case 1: Financial and Transaction Management

Pre-condition: User is authenticated and has existing transaction records in the database

Basic Flow:

1. The system visualizes the Dashboard with four summary cards, which are Current Balance, Monthly Income, Monthly Expenses, and Estimated Tax to Save (Gemini Pro's estimated value).
2. The Transaction Management Services retrieves the "Recent Transactions" log from the database, categorizing entries into 'Business' and 'Personal' purposes.
3. The user utilizes the "Split" Action to partition a single transaction into multiple cost centers. For example, Users can separate a bulk purchase into personal and business expenses.

Post-condition: The system stores all transaction data and split preferences in Google Firebase. When the user returns to the dashboard later, their data loads from the cloud automatically.

Use Case 2: Inventory evaluation and Predictive Budgeting

Pre-condition: Users have initialized inventory items and budget limits.

Basic Flow:

1. The user accesses the Inventory & Forecast Section to monitor stock levels at the Current Inventory Table. For example, if the user is a baker, they will check into Flour, Sugar, Butter, and Eggs.
2. The system triggers a Low-stock Item, which will flag the items that consist of quantities below a defined threshold with a visual color indicator.
3. The user can click on the Add Item button to log new inventory entries and click on Update that sits next to existing items to adjust quantities. These operations trigger an immediate recalculation of the total item values.
4. At the same time, the Budget Planner plots actual expenditure against predefined limits based on categories such as ingredients or even marketing using progress bars.
5. The user can see the Gemini Pro Forecast, which is an AI-powered prediction based on the prior 3 months of data. It shows estimated values of next month's revenue, expenses, and Tax Savings.

Post-condition: Gemini Forecast card displays AI-predicted revenue, expenses, and recommended tax savings for the next month, and users can click "Refresh" to recalculate based on the latest data input, and the system will automatically update the Budget Planner panel as well as Dashboards in Use Case 1.

Use Case 3: Conversational AI Analysis and Data Ingestion

Pre-condition: The system is actively connected to the Gemini Pro interference engine.

Basic Flow:

1. The user provides input via three modalities, which are through Document Upload

(CSV/PDF), Receipt Image (OCR processing), or even using Voice Input for hands-free data entry at the Upload & Analyze panel and later committed to the Transaction database.

2. The system parses the input to identify the user intent. For example, “Analyze Groceries Spending Wise.”
3. The AI Analyst cross-references the query with the user’s historical financial database.
4. The system provides financial recommendations or a summary of the uploaded document.
5. The users can navigate to the AI chat inference and type a question like “How much should I save for tax?” or select a Quick Question shortcut if the users are unsure of what to ask.

Post-condition: The interaction is logged in the chat history for future contextual retrieval.

5.3 Use Case Diagram:

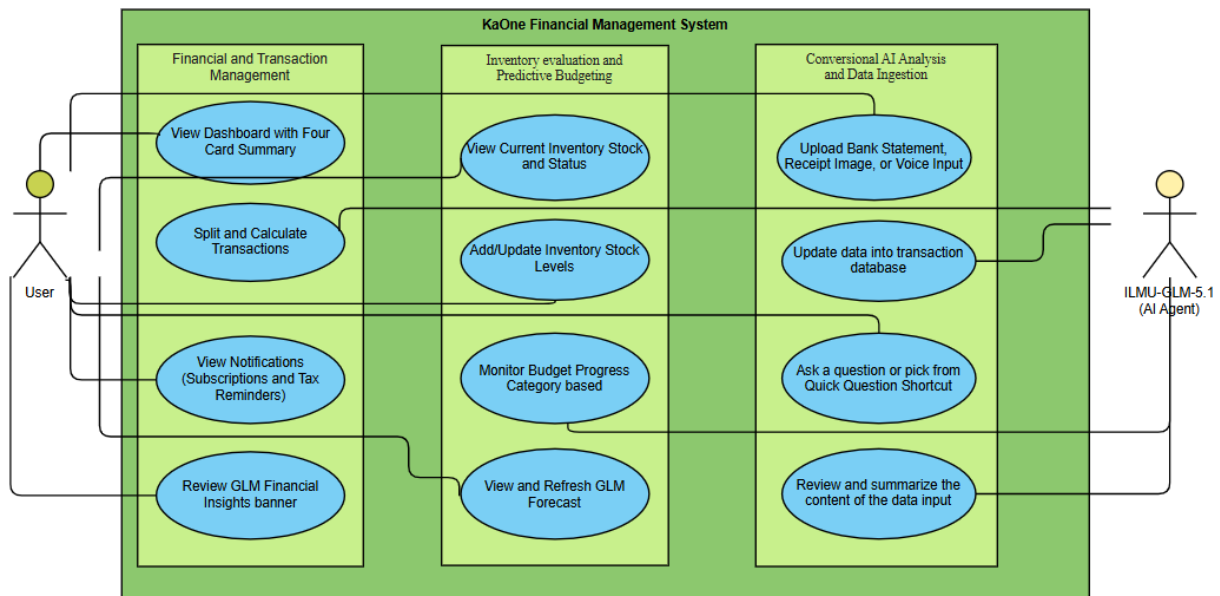


Figure 5.0 Use Case Diagram

6. Features Included (Scope Definition)

- **Interactive financial dashboard:** A conversational interface where users can manage inventory, budgets, and view AI-generated insights in a single 2D workspace.
- **Conversational AI Analyst for financial queries:** A chatbot where you can ask anything about financial using natural language.
- **Multi-Modal Input:** Support for Text-to-Finance (chat queries), Picture-to-Finance (receipt scanning and analysis), Voice-to-Finance (Voice input from users) and Document-to-Finance (Upload document such as CSV, PDF or Excel)
- **Gemini Pro-Powered Forecasts:** Automated prediction of next month's revenue, expenses and LHDN tax savings based on historical patterns.
- **Subscription Kill Switch:** An AI agent that identifies unused services and drafts cancellation messages
- **Smart transaction splitter:** A specialized tool to toggle transactions between "Personal" and "Business" categories for clean bookkeeping.
- **Cloud database sync using Firebase Firestore:** User data is saved in the cloud, not just in the browser. Users can log in from any device and see their full transaction history.

7. Features Not Included (Scope Control)

- 7.1. Real Bank API Integration

We don't connect directly to Maybank, CIMB, or any bank APIs. Users must upload CSV files or enter transactions manually. This keeps the app simple and avoids complex banking regulations for the hackathon.

- 7.2. Multi-User Support

Only one user per account. No team access, no shared dashboards, no role-based permissions. Each business owner manages their own data separately.

- **7.3. Mobile App**

Web browser only. No iOS or Android app for now. The dashboard works on phone browsers but it's not a native mobile experience.

- **7.4. Automatic Transaction Syncing**

Banks don't auto-push transactions to KaOne. Users must upload files or type in their expenses. No real-time syncing or scheduled imports.

- **7.5. Payment Processing**

You cannot pay bills or transfer money through KaOne. It gives advice and predictions only, no actual money movement.

- **7.6. Limited Security Auditing**

We use Firebase for cloud storage which provides standard security rules. However, we have not undergone third party security audits. For a hackathon prototype, this is acceptable. For production, we would add additional encryption and audit logging.

- **7.7. Offline Mode**

Internet connection required at all times. The AI calls need API access to Gemini Pro.

- **7.8. LHDN Official Filing**

KaOne predicts your tax but does NOT file taxes for you. You still need to submit your own LHDN forms. Consider this a calculator, not an official tax agent.

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate an average token cost of RM 0.05 to RM 0.15 per user session. To manage costs at scale, we implemented semantic caching for frequent queries such as general LHDN tax rate inquiries to avoid unnecessary model inference.
- **Technical Constraints:** Financial report exports are currently limited to PDF and CSV formats. Direct integration with Malaysian banking APIs such as Maybank, CIMB or auto-filing to the LHDN portal is not supported in the current version.

- **Performance Constraints:** There is a limit of 10 AI-generated forecasts per day for Freemium users to maintain high-fidelity tax and budget predictions,
- **User Input:** The model cannot proactively execute financial changes like cancelling a real subscription without explicit human approval through the "Confirm Cancel" button.

9. Risks & Questions Throughout Development

- **9.1 AI Hallucination Risk**

What happens when Gemini Pro gives wrong tax advice?

Our approach: Every AI response includes a disclaimer: "AI-generated suggestion. Verify with a licensed accountant for official tax filing." We also flag high-confidence predictions vs. low-confidence ones. If the model is uncertain, it says "I'm not sure — here's a general guideline instead."

- **9.2 Data Privacy Concern**

User uploads receipts with personal info. Where does it go?

We use Firebase Firestore to store transaction data in the cloud. This is necessary for users to access their data from multiple devices. Receipt images are processed and then discarded. Only extracted transaction text is stored. Users can delete their data at any time from the settings page.

- **9.3 API Key Exposure**

The ILMU API key is visible in the code.

For hackathon demo, this is fine because it's a free tier key with limited usage. For a real product, we would move the key to a backend server and never expose it to the browser.

- **9.4 Model Response Speed**

ILMU API calls take 2-5 seconds sometimes.

We add typing indicators and loading spinners so users know the AI is working. No one stares at a frozen screen wondering if it crashed.

- **9.5 Misclassified Transactions**

The Splitter learns from user input, but what if it learns wrong patterns?

We let users manually override any AI-suggested category. Each transaction keeps a history of changes so users can revert mistakes.

- **9.6 Subscription Kill Switch Accuracy**

How does the app know a subscription is "unused"?

We don't actually track usage — we rely on the user telling us, or we flag based on transaction age (if no payment in 30+ days). This is a demo simplification. A real version would need to integrate with subscription services directly.

- **9.7 Cross-Browser Compatibility**

We tested on Chrome and Safari. What about older browsers?

The app uses modern CSS (backdrop-filter, grid, flexbox). It will break on Internet Explorer and very old Android browsers. We add a warning banner for unsupported browsers.

- **9.8 Concurrent Users**

Since we use Firebase, each user's data is stored under their own account. There is no cross contamination between different users. Each judge sees only their own transactions and settings.

- **9.9 What If Gemini Pro Goes Down?**

API outage or rate limit hits.

We built fallback mock responses. The app still works and shows example answers, just not live AI. Judges can still see the flow and UI even if the API is slow.